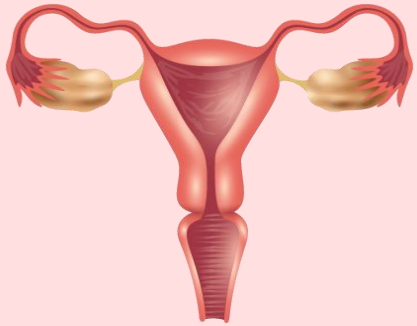
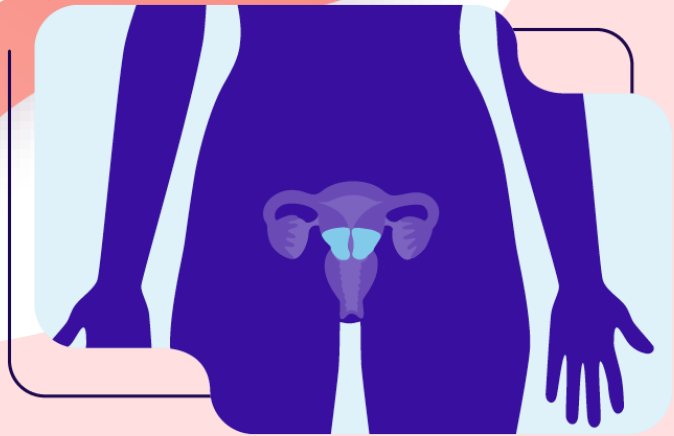
Multi-Level Fuzzy Inference System for Cervical Cancer Risk Assessment



Ana Maria Alvarez – DVOR2C

Machine Intelligence



The highest rates of cervical cancer incidence and mortality are in low- and middle-income countries.

4th

Most common cancer in women globally

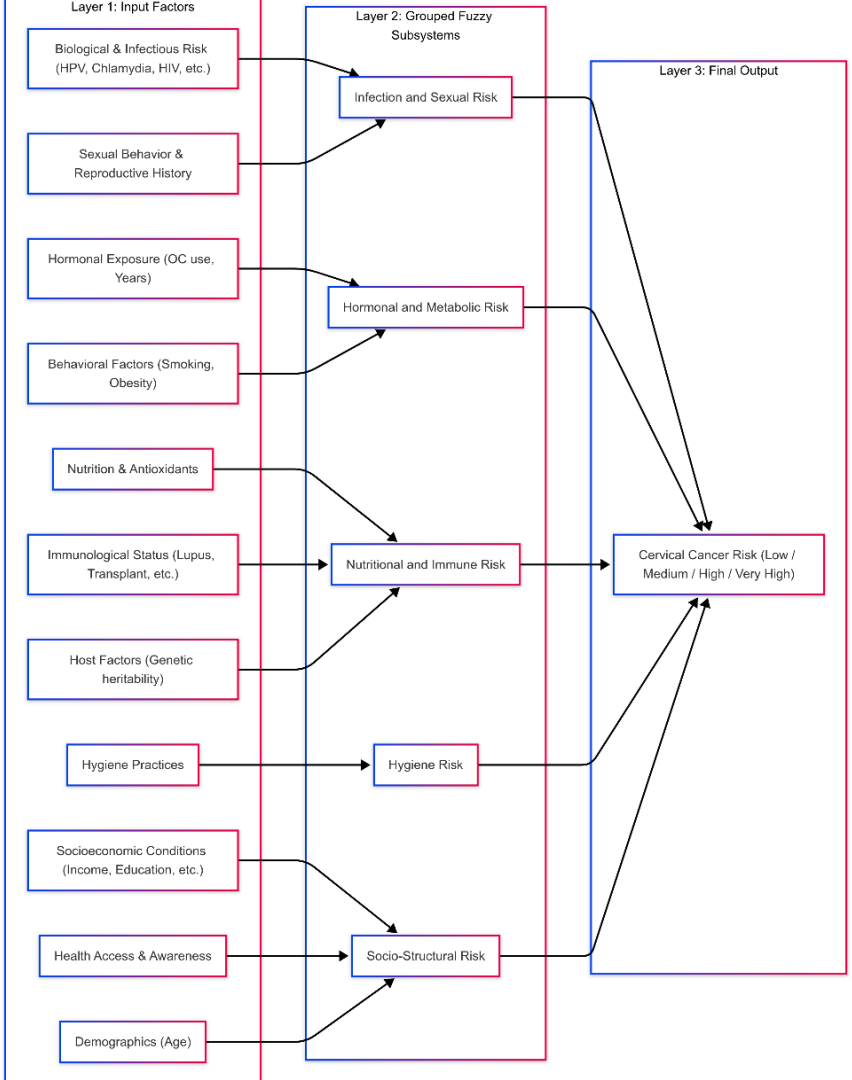
660,000

New cases in 2022

350,000

Deaths in 2022

Fuzzy System Design



Layer 1 – Classification of risk factors

Biological Infectous

- HPV
- HPV type
- Chlamydia
- HIV
- HSV
- Bacterial Vaginosis
- Genial warts

Sexual reproductive

- Number of sexual partners
- Age at first sexual activity
- Parity
- Age at first pregnancy

Hormonal exposure

- Use of Oral contraceptive pills
- Years using OC

Behavioral

- Years of smoking
- Packs per year
- Obesity

Hygiene practices

- Use of vaginal cleansers
- Bath history
- Bath during menstruation
- Material used during menstruation
- Wash after sexual intercourse

Nutritional

- Balanced diet
- Vitamin C
- Vitamin E
- Antioxidants

Layer 1 – Classification of risk factors

Immunological status

- Rheumatoid arthritis
- Systemic lupus erythematosus
- Use of immunosuppressive agents
- Organ transplant

Host factors

- Genetic heritability

Demographic

- Age

Socioeconomic

- Education
- Housing type
- Place of residence
- Marital status
- Monthly household income

Health access

- Health facility access
- Health services used
- Awareness of vaccination
- Awareness of pap smear

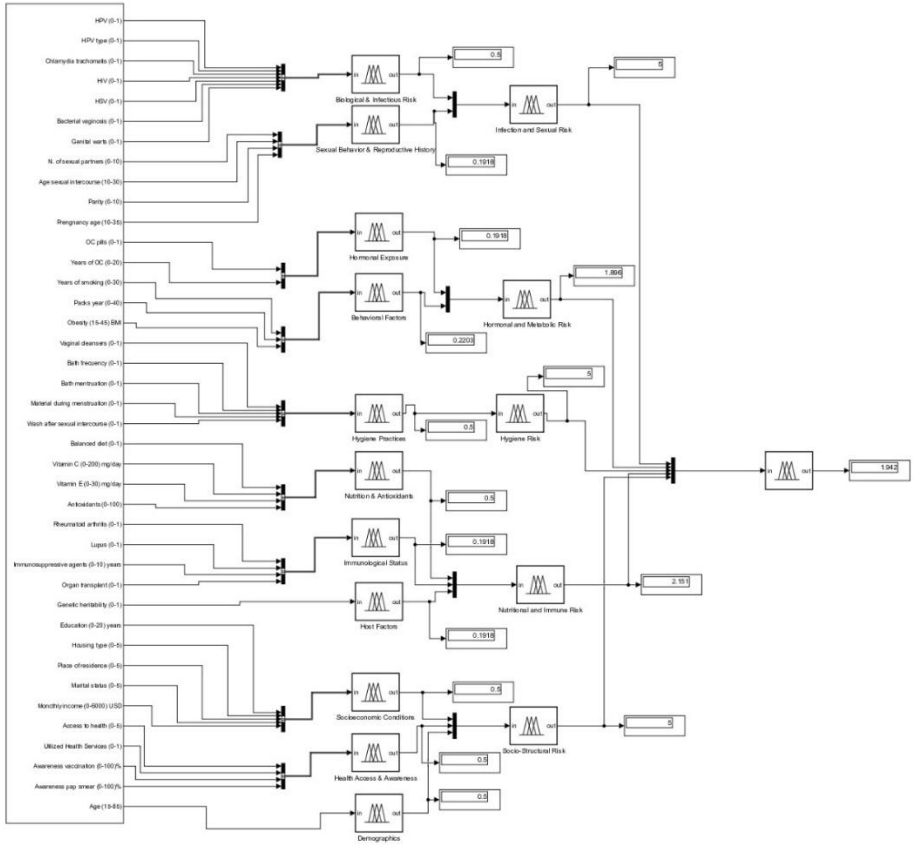
Layer 2 – Interpretable risks

- Infection and Sexual risk
- Hygiene risk
- Socio structural risk
- Nutritional an Immune risk
- Hormonal and Metabolic

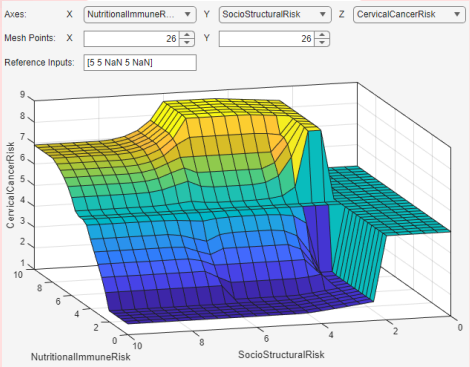
Layer 3 – Cervical Cancer Risk

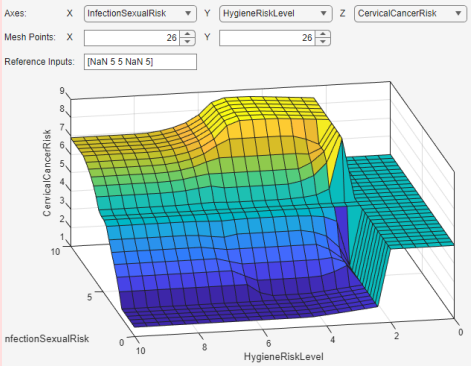
Input	Range	Low MF	Medium MF	High MF
InfectionSexualRisk	[0–10]	trapmf [0 0 2 4.5]	trimf [2.5 5 7.5]	trapmf [5.5 8 10 10]
HormonalMetabolicRisk	[0–10]	trapmf [0 0 2 4.5]	trimf [2.5 5 7.5]	trapmf [5.5 8 10 10]
NutritionalImmuneRisk	[0–10]	trapmf [0 0 2 4.5]	trimf [2.5 5 7.5]	trapmf [5.5 8 10 10]
HygieneRiskLevel	[0–10]	trapmf [0 0 2 4.5]	trimf [2.5 5 7.5]	trapmf [5.5 8 10 10]
SocioStructuralRisk	[0–10]	trapmf [0 0 2 4.5]	trimf [3 5 7]	trapmf [6 8 10 10]

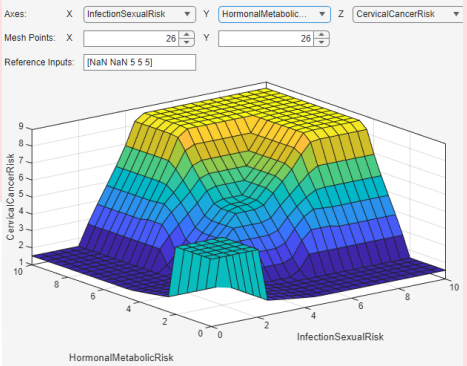
Results

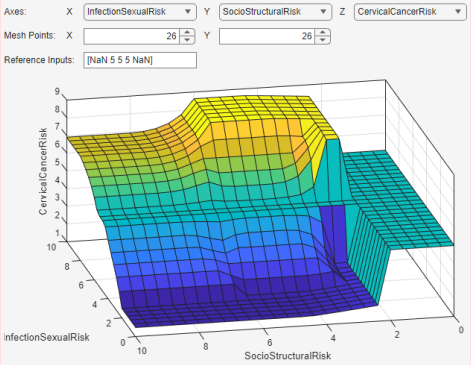

HPV (0-1)
HPV type (0-1)
Chlamydia trachomatis (0-1)
HIV (0-1)
HSV (0-1)
Bacterial vaginosis (0-1)
Genital warts (0-1)
N. of sexual partners (0-10)
Age sexual intercourse (10-30)
Parity (0-10)
Pregnancy age (10-35)
OC pills (0-1)
Years of OC (0-20)
Years of smoking (0-30)
Packs year (0-40)
Obesity (15-45) BMI
Vaginal cleansers (0-1)
Bath frequency (0-1)
Bath menstruation (0-1)
Material during menstruation (0-1)
Wash after sexual intercourse (0-1)
Balanced diet (0-1)
Vitamin C (0-200) mg/day
Vitamin E (0-30) mg/day
Antioxidants (0-100)
Rheumatoid arthritis (0-1)
Lupus (0-1)
Immunosuppressive agents (0-10) years
Organ transplant (0-1)
Genetic heritability (0-1)
Education (0-23) years
Housing type (0-5)
Place of residence (0-5)
Marital status (0-5)
Monthly income (0-6000) USD
Access to health (0-5)
Utilized Health Services (0-1)
Awareness vaccination (0-100)%
Awareness pap smear (0-100)%
Age (15-85)
Biological & Infectious Risk
Sexual Behavior & Reproductive History
Infection and Sexual Risk
Hormonal Exposure
Behavioral Factors
Hormonal and Metastatic Risk
Hygiene Practices
Hygiene Risk
Nutrition & Antioxidants
Immunological Status
Host Factors
Nutritional and Immune Risk
Socioeconomic Conditions
Health Access & Awareness
Socio-Structural Risk
Demographics
0.5
0.1918
0
0.1918
1.896
0.2201
5
0.5
0.5
0.1918
0.1918
2.151
0.5
0.5
0
0.5
1.942

Results


Axes: X NutritionalImmuneR... Y SocioStructuralRisk Z CervicalCancerRisk
Mesh Points: X 26 Y 26
Reference Inputs: [5 5 NaN 5 NaN]
CervicalCancerRisk
NutritionalImmuneRisk
SocioStructuralRisk


Axes: X InfectionSexualRisk Y HygieneRiskLevel Z CervicalCancerRisk
Mesh Points: X 26 Y 26
Reference Inputs: [NaN 5 5 NaN 5]
CervicalCancerRisk
nfectionSexualRisk
HygieneRiskLevel


Axes: X InfectionSexualRisk Y HormonalMetabolic... Z CervicalCancerRisk
Mesh Points: X 26 Y 26
Reference Inputs: [NaN NaN 5 5 5]
CervicalCancerRisk
HormonalMetabolicRisk
InfectionSexualRisk


Axes: X InfectionSexualRisk Y SocioStructuralRisk Z CervicalCancerRisk
Mesh Points: X 26 Y 26
Reference Inputs: [NaN 5 5 5 NaN]
CervicalCancerRisk
nfectionSexualRisk
SocioStructuralRisk

Resources

"Basic Information about HPV and Cancer," CDC - cancer, 17 September 2024. [Online]. Available: https://www.cdc.gov/cancer/hpv/basic-information.html#:~:text=High%2Drisk%20and%20low%2Drisk%20HPV%20types&text=The%20types%20of%20HPV%20that,and%2060%25%20of%20penile%20cancers.. [Accessed April 2025].
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"Risk Factors for Cervical Cancer," American Cancer Society, 3 January 2020. [Online]. Available: https://www.cancer.org/cancer/types/cervical-cancer/causes-risks-prevention/risk-factors.html. [Accessed 2025].
L. Mirabello and W. N. Chase, "Human papillomavirus genomics: Understanding carcinogenicity," Tumour Virus Research, vol. 15, June 2023.